PROJECT SILO TECHNICAL MEMO #12

DESIGN CONSIDERATIONS AND PRELIMINARY MEASUREMENTS OF CORE DRIVER TRANSISTORS

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ABSTRACT: Some of the design considerations for the development of a core driver transistor for a 0.5 microsecond core memory are discussed. Preliminary data on germanium graded base transistors indicate that an NPN device can be made to switch one ampere in less than 50 millimicroseconds. This device will probably have a collector breakdown voltage in the range between 100 and 130 volts. The transistor should be able to dissipate at least 7 watts if sufficient heat sink cooling is provided.

If one desires a device which will tolerate collector voltages in the range between 300 and 400 volts it is necessary to consider the use of silicon for this purpose.

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I. Introduction

A program is being carried out to determine the feasibility of using transistor drivers for a half microsecond core memory. The core switching requirements for this transistor can be summarized as follows:

1. Ability to turn on one ampere of collector current in less than 50 millimicroseconds,
2. Ability to turn off one ampere of collector current in less than 50 millimicroseconds,
3. Ability to withstand collector voltages up to 400 volts, and
4. Ability to operate at about 7 watts dissipation.

In the initial phases of this feasibility study, one can state that any one of these four conditions can be obtained in an individual device. It will require a considerable research and development effort to be able to realize all four conditions in an individual device.

In section II a discussion of the following will be given:

1. Power dissipation
2. Current limitations
3. Speed
4. Voltage breakdown

II. Discussion of Design Considerations

A. Power Dissipation

In order to obtain a value of transistor power dissipation under typical use the following relation was used*;

$$P_{ave} = \frac{V_c I_m}{6} \frac{t_1}{\tau} + \frac{I_m V_{cs} t_2}{\tau} + \frac{V_c I_m t_3}{6 \tau} + \frac{I_{co} V_c t_4}{\tau}$$

where the following values when substituted

τ = the period = 500×10^{-9} seconds

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* This relation was derived by integration of the instantaneous power over the period τ. The major assumption used is that the transition from off to the on state is linear.

t_1 = turn-on transition time = 50×10^{-9} seconds

t_2 = the conduction time = 200×10^{-9} seconds

t_3 = turn-off transition time = 50×10^{-9} seconds

t_4 = off time = 200×10^{-9} seconds

I_m = full conduction current = 1 ampere

I_{co} = off current = 10^{-3} amperes

V_{cs} = collector saturation voltage (at I_m = 1 ampere)= 1V

V_c = collector voltage in off state = 200 volts

yield

P_{ave} = 7.14 watts.

In designing a transistor to dissipate this quantity of power, one can separate the problem into two parts; (1) heat transfer from junction to heat sink, and(2) heat extraction from the heat sink.

In this early stage of development, heat transfer to the sink is the prime concern. If one assumes that the collector junction should not rise more than 30°C above sink temperature, then the "K" factor of such a device must be about 4.2°C/watt.

The average drift transistor structure as made by the fabrication group has a "K" factor of 700°C/watt. By attachment of a copper base tab to a heat sink, it has been possible to reduce the "K" factor to 50°C/watt. By attachment of the collector directly to a heat sink it has been possible to obtain a "K" factor of 8°C/watt. By further reducing the thickness of collector material to less than half of its present value, one should be able to reduce the "K" factor to less than 4°C/watt.

B. Current Limitations

The calculation of the average power dissipation in Section A was made using a V_{cs} of one volt. The present IBM drift transistor has a V_{cs} (at I_c = 1 ampere) between 10 and 20 volts. By reducing the thickness of the collector region it has been possible to reduce the saturation resistance to about one ohm thereby reducing V_{cs} to about one volt at one ampere.

As one increases the emitter current the large-signal alpha prime passes through a maximum and then falls off. Alpha falloff occurs when the injected minority carrier density approaches and exceeds the majority carrier density of the base region. In order to have a useful device it is desirable to have a large-signal current gain greater than about 10. One method of decreasing the rate of alpha falloff is to increase the emitter area. An increase in emitter area decreases the current density for a given current. Alpha falloff is thereby diminished. Such an increase cannot be carried out without limit because as the emitter area is increased the emitter-to-base transition capacitance increases. This causes an increase in the input RC time constant of the device which lowers the frequency response of the transistor.

The effect of alpha falloff can also be reduced by using impurities in the emitter alloy which have high segregation coefficients. This step increases the conductivity of the emitter region and therefore increases emitter efficiency over a larger range of injected current density.

NPN transistors have a more favorable alpha falloff characteristic than PNP transistors because of the mobility difference between electrons and holes.

C. Speed

There are a number of factors that determine the speed of a transistor. The fundamental limit in speed of a transistor is the transit time of minority carriers across the base region. By taking advantage of the built in sweeping field of a graded base transistor, one can reduce this time to a minimum. In addition to transit time delays, a transistor may also be limited by the RC time constants of both collector and emitter junctions. The speed of the core driver transistor is primarily limited by the emitter capacitance. As was pointed out in the last section, a compromise in emitter size must be made in order to try to obtain the required current at the desired rate of operation. At the present time it appears that an emitter junction of about 0.010" in diameter is optimum. It might be possible to increase emitter performance by depositing an intrinsic layer between the emitter junction and the base region. This would serve to decrease the emitter junction capacitance. It might then be possible to increase the emitter area without a sacrifice in speed.

D. Voltage Breakdown

The requirement of a collector voltage breakdown above 400 volts is by far the most difficult to meet. Theoretically a breakdown voltage of this magnitude or higher should be possible to obtain in germanium. In actual practice the highest breakdown observed in this laboratory for germanium diffused junctions is about 200 volts. Breakdown of these junctions generally exists between 100 and 150 volts. The deviation from theory can apparently be attributed to surface breakover effects.

Breakdown voltages of about 400 volts have been observed in this laboratory for relatively high resistivity silicon alloy junctions. However, as yet, a satisfactory silicon transistor has not been made.

III. Measurement Techniques

Improving the design of a transistor requires sufficient knowledge of the parameters of the existing device so that suitable changes in these parameters can be made to achieve the desired characteristics. This knowledge includes information concerning the static characteristics, switching times, voltage breakdown, and "K" factor (heat dissipation). For the core driver transistor, obtaining this information presents problems in instrumentation due to high power dissipation at high current levels.

Measuring static characteristics of the core driver transistor by conventional d-c plotting methods requires that the unit be able to dissipate average powers of the order of 20 watts. Because of this, a pulse sampling technique was devised to allow the testing of low power dissipation units at the high peak power levels. This technique consists of turning the unit on for a period of about 5 μsecs., making the measurement in this time and then turning the device off for a cooling period which is many times longer than the on time. If this duty cycle is made to be a ratio of 1/1000, a 100 mw unit can safely dissipate a peak power of 10-20 watts. Families of static characteristics of collector current vs. collector voltage and base current, plus curves of base voltage vs. collector voltage and base current were obtained using this technique. From these characteristics, information such as small signal α', large signal α', saturation resistance, dynamic collector resistance, input resistance, etc. can be obtained at the high power dissipation levels.

Switching time is obtained by the standard method of applying an input pulse and observing the output pulse on an oscilloscope having sufficient rise and fall time capabilities. Since this is done at a low duty factor, switching time at high current, high power levels can be observed.

At present λ frequency cutoff and/or λ' frequency cutoff are not measured due to the high power dissipation encountered with continuous operation.

Collector voltage breakdown is an important characteristic in this device since the core driver transistor is required to withstand high voltages in its application. The breakdown voltage is obtained by observing on an oscilloscope the plot of collector to base diode back current vs. collector to base voltage. Collector breakdown voltage is defined as the collector to base voltage required to produce 500 μamps back current.

Power dissipation capabilities, or "K" factor measurements, are made by measuring the electrical power consumed to produce a rise in junction temperature. In this case "K" factor is defined as the change in junction temperature relative to case temperature divided by the power dissipation required to produce this change in temperature. The details of the actual measurement are as follows:

1. An X-Y recorder is used to obtain a plot of the back-current characteristics of the collector-to-base junction vs. temperature as the temperature is varied above ambient.

2. The transistor is placed in a circuit which allows electrical power to be applied to the device. The case of the transistor is immersed in Freon 11, which maintains the case temperature at 23.8°C due to the boiling action of the Freon 11. After the power has been applied, the power is momentarily cut off and immediately thereafter the back current of the collector to base junction is measured with the aid of an oscilloscope. This step is repeated for a number of different values of power input.

3. Knowing the collector to base junction back current at a given power dissipation, the junction temperature can be obtained from the information obtained in step 1.

4. A plot of junction temperature vs. power dissipation is made, the slope of this line being the "K" factor of the transistor.

IV. Discussion of Experimental Devices

A. Saturation Resistance

In the early stages of this development the standard switching type of drift transistor was scaled up in size to handle the higher current requirements of the core driver. The structure consists of a diffused collector junction, graded base, and alloy emitter junction. The collector junction was a 0.060" diameter diffused junction. The emitter alloy region had a diameter of about 0.010". A ring type of base tab having a 0.025" hole was soldered to the base layer. The thickness of germanium in the collector ranges from 0.004" to 0.005" for these units. Figure 1 shows a plot of collector current vs. collector voltage at several different base current levels (grounded emitter operation) for a typical structure of this design. The collector to emitter resistance is about 20 ohms for a collector current of 125 ma. Since it is desirable to keep the collector to emitter voltage below one volt, a change in structure was necessary.

The first step toward lowering the saturation resistance was to decrease the thickness of germanium on the collector side of the diffused junction. This was done by carefully lapping and preferentially etching the diffused germanium wafer before it was cut into discs. The first units made in this manner showed considerable improvement as evidenced by the typical unit shown in figure 2. The collector to emitter saturation resistance is about 2 ohms. Further reduction in this resistance value can be achieved by using a jet etching technique on the collector side of the junction so that ohmic contact can be made at a distance of about 0.0002" from the collector junction. This method is presently being evaluated.

Another method of reducing the saturation resistance is to use an alloy junction collector in place of the diffused junction collector. The first attempt in this direction gave the results shown in figure 3. The saturation resistance of this device is definitely down in the desirable range. This method, however, does have certain disadvantages, namely:

1. Since two alloy junctions must be used it is doubly difficult to control base width,

2. Collector capacitance per unit area will be higher than for the diffused case,

3. Breakdown voltage may be lower than for the diffused case.

More work must be done before the decision can be made as to which solution to the saturation resistance problem is most feasible.

B. Speed

When discussing transition speed of an experimental transistor, one naturally uses the transition speed of a more common transistor as a comparison. In this case, the transition speeds of the IBM 01 transistor for PNP's and the IBM 51 transistor for NPN's are used as standards of comparison. Figures 4 and 5 show the transition characteristics of typical 01's and 51's respectively. In these figures, collector current is plotted vertically and time horizontally with the device operating in the circuit of figure 6. The transition speeds for these two units fall in the vicinity of 1-2 μ sec. for both turn-on time and turn-off time.

Figures 7 and 8 show the transition speeds for experimental graded base core driver transistors, unit #N71 (PNP) and unit #V458 (NPN), operating in the circuit of figure 6. The transition speeds for these units are about 0.5 μ sec. for the PNP and 0.2 μ sec. for the NPN. By comparison then, these experimental units are between 5 and 10 times faster than the standard IBM 01 and 51 types of transistors.

The speed of both standard and experimental models of the transistors can be improved by external circuitry. The capacitor shown dotted in figure 6 is one technique of speed-up circuitry used to demonstrate the ultimate speed capabilities of these units. Figures 9 and 10 show the transition response of the IBM type 01 and 51 respectively with an 800 $\mu\mu$fd capacitor used for the type 01 and a 750 uufd capacitor used for the type 51. These figures show that an improvement in transition speed by a factor of about ten is achieved by this technique. Figures 11 and 12 demonstrate the speed response of experimental units #N71 and #V458 with similar speed-up circuitry. In the case of N71 a 400 $\mu\mu$fd capacitor was used. In all cases, the transistors were driven to a collector current level of 100 ma. with the scope sensitivity at 25 ma/cm.

From the speed point of view, it can be said that the graded base units are between five and ten times faster than non-graded base units, and when speed-up circuitry is used, the degree of speed-up circuitry required for the graded base units is less than that required for the non-graded base units.

C. Large Signal α'.

Large signal α' is an important consideration in the core driver transistor because it is the direct measure of the static current required to drive the device to the desired one ampere collector current level. No definite value has been set for large signal α' in this application but a reasonable figure to consider would be a value between ten and twenty.

Again using the IBM types 01 and 51 in comparison to the experimental types, figures 13, 14, 15, and 2 show that "α" crowding is more serious in the PNP than in the NPN. This indicates that the desired values of large signal α' for a collector current of one ampere can be achieved more readily with the NPN type of transistor. Figure 2 is a plot of I_c vs. V_c for an experimental graded base NPN unit #N84 which shows a large signal α' of about 35 at a collector current level of 550 ma. Though this unit was not driven to the one ampere level, the large signal α' at 550 ma. indicates that a large signal α' at the one ampere level should lie in the range between 10 and 20.

V. Conclusions

The data lead one to believe that, with the present state of the art, an NPN germanium transitor can be designed with the following characteristics:

Maximum collector breakdown $\sim$ 100-130 volts

Switching time with grounded emitter
(without speed-up circuitry) $\sim$ 0.2 μ sec.
(with speed-up circuitry) $\sim$ 0.05 μ sec.

Collector back current at 24°C $\leq$ 100 μ amps.

Input voltage, grounded emitter(I_c=1 amp) $\sim$ 0.6-1.0 volts

Average current gain (I_c from 0 to 1 amp) $\sim$ 10-20

Emitter breakdown voltage $\sim$ 2-8 volts

Punch-thru voltage $>$ collector breakdown voltage

Average power dissipation to an infinite heat sink at 24°C $>$ 7 watts

Minimum collector to emitter voltage (I_c=1 amp) $\sim$ 0.5-1.0 V

Maximum collector current $\sim$ 1 ampere

The only characteristic which does not meet the requirements set forth in the introduction is collector breakdown voltage. If it is imperative to increase collector breakdown to 400 volts, it is necessary to intensify the existing effort on silicon device research. The step is necessary because present data indicate that a breakdown of 400 volts can be obtained in silicon but not in germanium. Some of the advantages and disadvantages of using silicon are listed below:

Advantages

1. Higher breakdown voltages are possible.
2. Junctions can tolerate higher temperatures thereby simplifying cooling problems.
3. Junction transition capacitance/unit area should be less, since the dielectric constant for silicon is 12 compared to 16 for germanium.

Disadvantages

1. Speed may be decreased by the smaller carrier mobility found in silicon.
2. The minimum collector to emitter voltage at one ampere collector current will be increased by approximately 0.4 volts.
3. The input voltage to turn the device on will be increased by about 0.4 volts.
4. From a time schedule point of view there may be considerably more delay in the development of a silicon device, because silicon technology is not as advanced as that of germanium.

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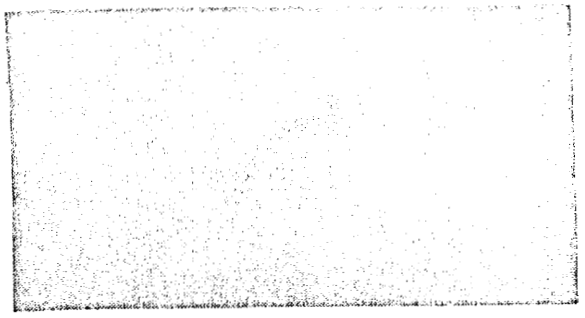


Figure 1

NPN # V450

Vertical....I_c = 125 ma/cm
Horizontal..V_c = 5 V/cm

I_b step = 2 ma
except top curve which is 3.2 ma

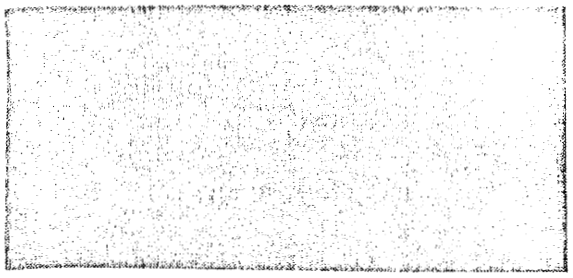


Figure 2

NPN # N84

Vertical....I_c = 125 ma/cm
Horizontal..V_c = 5V/cm

I_b step = 2 ma

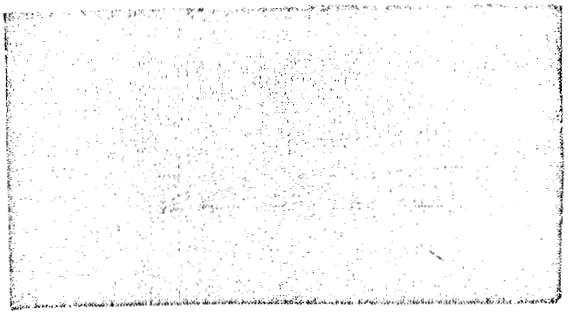


Figure 3

PNP # N71

Vertical....I_c = 0.2 ampere/cm
Horizontal..V_c = 1 V/cm

I_b not constant

#1 top curve I_b = 150 ma
2 I_b = 46 ma
3 I_b = 16 ma
4 I_b = 8 ma

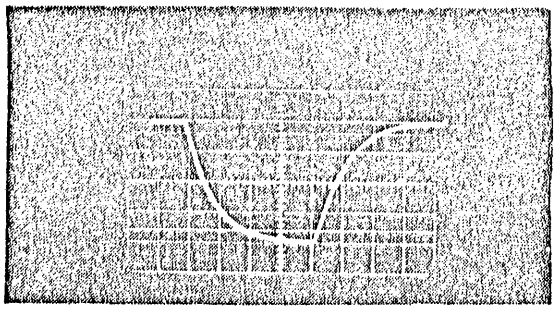


Figure 4

Plant PNP-01

Vertical....I_c = 25 ma/cm
Horizontal..Time = 1 usec/cm

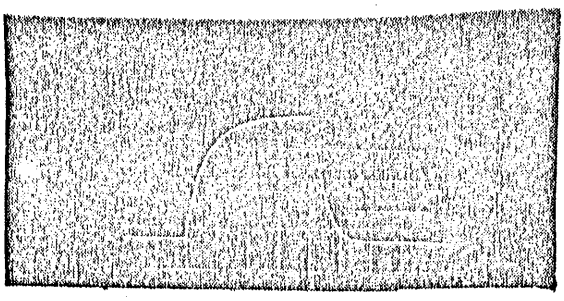


Figure 5

Plant NPN

Alloy Emitter Collector

Vertical....I_c = 25 ma/cm
Horizontal..Time = 1 usec/cm

Average D.C. collector current adjusted to 2 ma

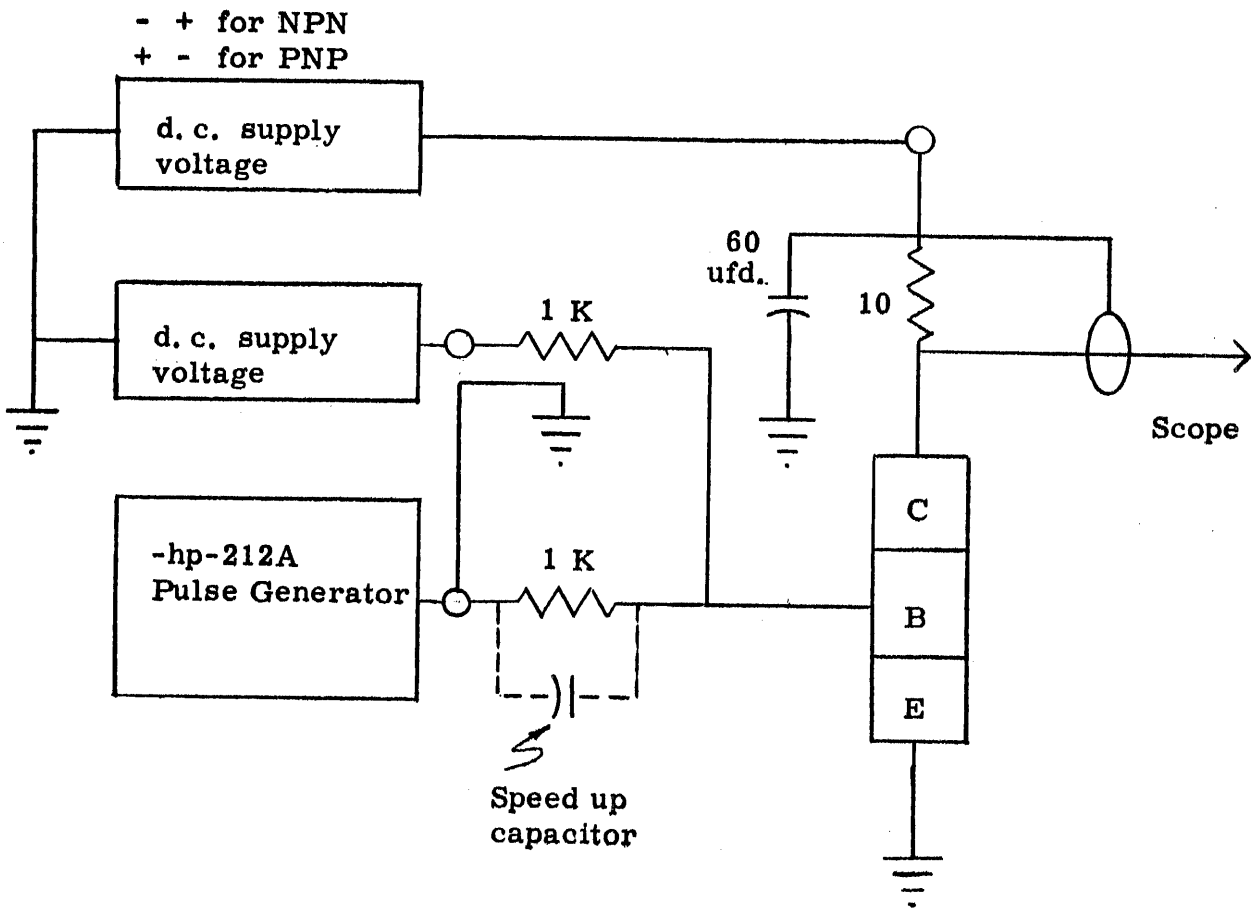

- + for NPN
+ - for PNP
d. c. supply voltage
d. c. supply voltage
-hp-212A Pulse Generator
1 K
1 K
60 ufd.
10
Scope
C
B
E
Speed up capacitor

Figure 6

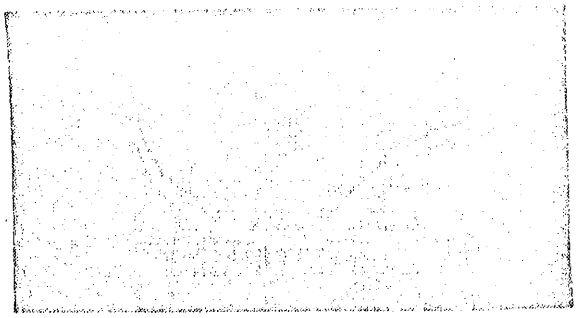


Figure 7

PNP # N71

Vertical....I_c = 25 ma/cm
Horizontal..Time = 0.2 μsec/cm

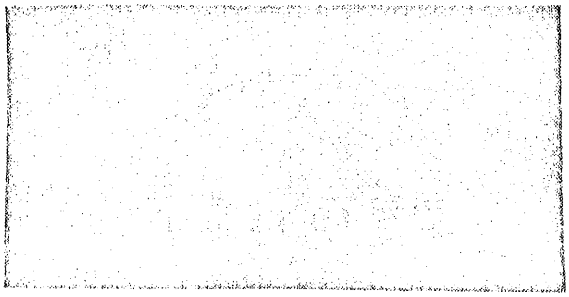


Figure 8

NPN # V458

Vertical....I_c = 25 ma/cm
Horizontal..Time = 0.1 μsec/cm

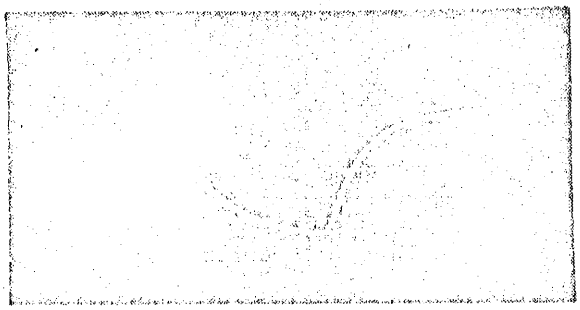


Figure 9

Plant PNP-01

Vertical....I_c = 25 ma/cm
Horizontal..Time = 0.1 μsec/cm

800 uufd speed up capacitor

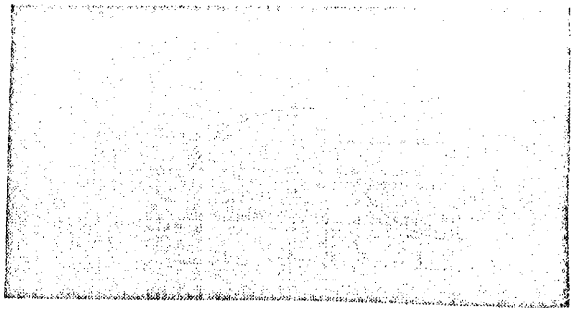


Figure 10

Plant NPN 51

Vertical....I_c = 25 ma/cm
Horizontal..Time = 0.1 μsec/cm

750 uufd speed up capacitor

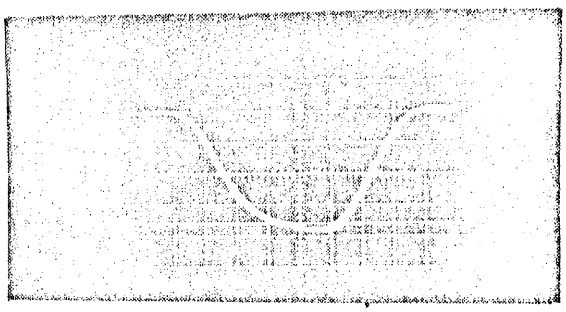


Figure 11

PNP # N71

Vertical....I_c = 25 ma/cm
Horizontal..Time = 0.02 μsec/cm

Speed up capacitor = 400 μμfd

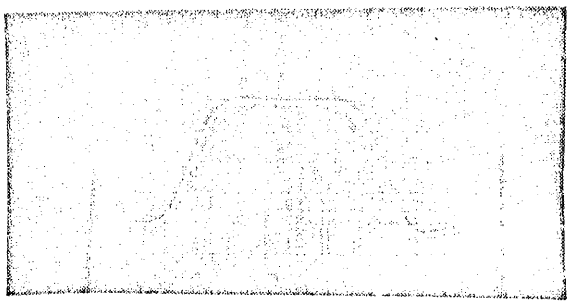


Figure 12

NPN # V458

Vertical....I_c = 25 ma/cm
Horizontal..Time = 0.02 μsed/cm

Speed up capacitor = 100 μμfd

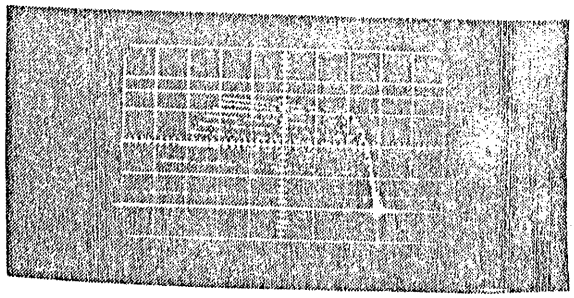


Figure 13

PNP TB4-01

Vertical....I_c = 250 ma/cm
Horizontal..V_c = 1 V/cm

ΔI_b = 10 ma

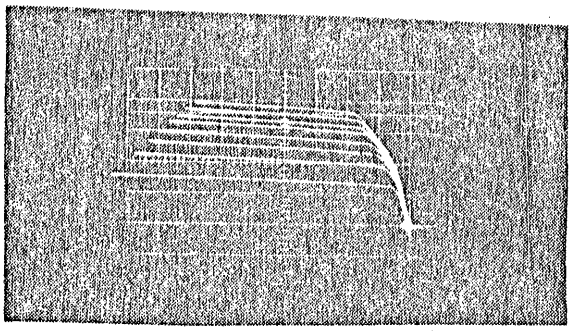


Figure 14

NPN IBM 51

Vertical....I_c = 0.5 ampere/cm
Horizontal..V_c = 1 V/cm

ΔI_b = 10 ma

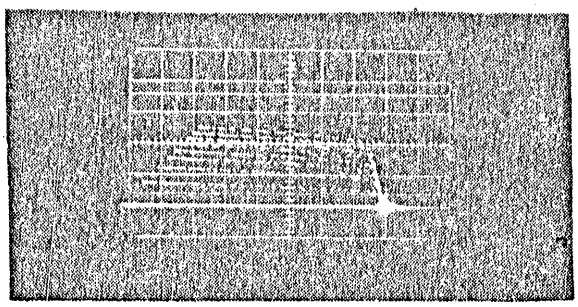


Figure 15

PNP 171

Vertical....I_c = [illegible] ma/cm
Horizontal..V_c = 1 V/cm

ΔI_b = 10 ma